19 Lecture 19

19.1 Complex numbers and integrals of complex valued functions

We recall the set of complex numbers $\mathbb{C} = \{x + iy : \ x, y \in \mathbb{R}\}$, where i satisfies $i^2 = -1$. For a complex number $z = x + iy$, we denote by $\mathrm{Re}(z) = x$ its real part, and by $\mathrm{Im}(z) = y$ its imaginary part. Its conjugate $\overline{z}$ is the complex number $x - iy$.

Sometimes it is useful to work with the *polar form* of a complex number: $z = r(\cos(\theta) + i\sin(\theta))$, where $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$. We also use the notation $|z| = r = \sqrt{x^2 + y^2}$.

We define the exponential function of a complex number $z \in \mathbb{C}$ as

$$e^z = \sum_{n=0}^{+\infty} \frac{z^n}{n!}.$$

Let's set $z = i\theta$ for some $\theta \in [0, 2\pi)$. We get

$$e^{i\theta} = \sum_{n=0}^{+\infty} \frac{i^n \theta^n}{n!} = \sum_{n \text{ even}} (-1)^{n/2} \frac{\theta^n}{n!} + i \sum_{n \text{ odd}} (-1)^{(n-1)/2} \frac{\theta^n}{n!} = \cos(\theta) + i\sin(\theta), \qquad \text{(Euler's identity)}$$

where we used the property $i^2 = -1$ and the Taylor expansions of $\cos(\theta)$, $\sin(\theta)$. We observe that

$$|e^{i\theta}| = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = 1.$$

For a complex valued function $f \colon \mathbb{R} \to \mathbb{C}$, we define its complex integral

$$\int f(x)dx := \int \mathrm{Re}(f(x))dx + i \int \mathrm{Im}(f(x))dx,$$

whenever the integrals on the right hand side exist.

Proposition 19.1 (Triangle inequality)**.** *Let $f \colon \mathbb{R} \to \mathbb{C}$ such that $\int f(x)dx$ exists. Then,*

$$\left| \int f(x)ds \right| \leq \int |f(x)|dx.$$

Proof. We write $\int f(x)dx$ in polar form: $\int f(x)dx = r(\cos(\theta) + i\sin(\theta)) = re^{i\theta}$, where we used Euler's identity. We have

$$\begin{aligned} r = \left| \int f(x)dx \right| &= e^{-i\theta} \int f(x)dx = \int e^{-i\theta} f(x)dx = \int \mathrm{Re}(e^{-i\theta} f(x))dx \\ &\leq \int |e^{-i\theta} f(x)|dx = \int |e^{-i\theta}||f(x)|dx = \int |f(x)|dx, \end{aligned}$$

and the result follows. The 4th equality holds because $\int e^{-i\theta} f(x)dx = \int \mathrm{Re}(e^{-i\theta} f(x))dx + i \int \mathrm{Im}(e^{-i\theta} f(x))dx$ is a real number, so $\int \mathrm{Im}(e^{-i\theta} f(x))dx = 0$. The inequality holds $\square$

19.2 Characteristic functions

Definition 19.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X a random variable with values in $\mathbb{R}$. The function $\phi_X \colon \mathbb{R} \to \mathbb{C}$ with

$$\phi_X(t) = \mathbb{E}[e^{itX}]$$

is called *characteristic function* of X.

Proposition 19.3. *Let X be random variable with values in $\mathbb{R}$. Then,*

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1. $|\phi_X(t)| \leq 1$.
2. $\phi_X(0) = 1$.
3. ϕ_X *is continuous.*

Proof. 1. Follows from Proposition 19.1.

2. We set $t = 0$ to find $\phi_X(0) = \mathbb{E}[e^0] = \mathbb{E}[1] = 1$.
3. For any $t \in \mathbb{R}$ we have

$$\begin{aligned}\lim_{h\to 0} |\phi_X(t+h) - \phi_X(t)| &= \lim_{h\to 0} |\mathbb{E}[e^{(t+h)iX} - e^{itX}]| = \lim_{h\to 0} |\mathbb{E}[e^{itX}(e^{ihX} - 1)]| \\ &\leq \lim_{h\to 0} \mathbb{E}[|e^{itX}||e^{ihX} - 1|] = \lim_{h\to 0} \mathbb{E}[|e^{ihX} - 1|].\end{aligned}$$

However, $|e^{ihX} - 1|$ is bounded because $|e^{ihX} - 1| \leq |e^{ihX}| + 1 = 2$, therefore by the Dominated Convergence Theorem we get

$$\lim_{h\to 0} |\phi_X(t+h) - \phi_X(t)| \leq \mathbb{E}[|e^0 - 1|] = 0.$$

□

Proposition 19.4 (Properties of characteristic functions)**.** *Assume that X, Y are two independent random variables and $a, b \in \mathbb{R}$.*

1. $\phi_X(-t) = \overline{\phi_X(t)}$, *where* $\overline{\phi_X(t)}$ *is the conjugate of* $\phi_X(t)$.
2. $\phi_{aX+b}(t) = e^{itb}\phi_X(at)$.
3. $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$.

Proof. 1. We have $\phi_X(-t) = \mathbb{E}[e^{-itX}] = \mathbb{E}[\overline{e^{itX}}] = \overline{\mathbb{E}[e^{itX}]} = \overline{\phi_X(t)}$.

2. $\mathbb{E}[e^{it(aX+b)}] = \mathbb{E}[e^{itb}e^{itaX}] = e^{itb}\mathbb{E}[e^{itaX}] = e^{itb}\phi_X(t)$.
3. By the independence, $\phi_{X+Y}(t) = \mathbb{E}[e^{itX}e^{itY}] = \mathbb{E}[e^{itX}]\mathbb{E}[e^{itY}] = \phi_X(t)\phi_Y(t)$.

□

Proposition 19.5 (Characteristic functions of known distributions)**.** *Let X be a random variable.*

1. *If* $X \sim Bin(n, p)$, *then* $\phi_X(t) = (pe^{it} + 1 - p)^n$.
2. *If* $X \sim Pois(\lambda)$, *for some* $\lambda > 0$, *then* $\phi_X(t) = e^{\lambda(e^{it}-1)}$.
3. *If* $X \sim Unif(-a, a)$, *for some* $a > 0$, *then* $\phi_X(t) = \begin{cases} \frac{\sin(at)}{at}, & \text{if } t \in \mathbb{R} \setminus \{0\}, \\ 1, & \text{if } t = 0. \end{cases}$
4. *If* $X \sim N(0, 1)$, *then* $\phi_X(t) = e^{-t^2/2}$. *In addition, if* $X \sim N(\mu, \sigma^2)$, *then* $\phi_X(t) = e^{it\mu}e^{-t^2\frac{\sigma^2}{2}}$.
5. *If* $X \sim Exp(\lambda)$, *for some* $\lambda > 0$, *then* $\phi_X(t) = \frac{\lambda}{\lambda - it}$.
6. *If* $X \sim \Gamma(a, \lambda)$, *for some* $a, \lambda > 0$, *then* $\phi_X(t) = \left(\frac{\lambda}{\lambda - it}\right)^a$.

Proof. We covered some calculations in class. □

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19.2.1 Characteristic functions and distributions

We observe that the characteristic function of a random variable X can be written as

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \int e^{itx}\, \mathbb{P}^X(dx).$$

In this subsection, we will see that knowing the characteristic function allows us to derive the probability distribution $\mathbb{P}^X$.

Roughly speaking, the characteristic function is the Fourier transform of the probability measure $\mathbb{P}^X$ (compare with the Fourier transform of a function), therefore ϕ_X defines $\mathbb{P}^X$ through an *inversion formula*.

Theorem 19.6. *Let X be a random variable and $\mathbb{P}^X$ be its distribution. Recall that $\mathbb{P}^X$ is a probability measure over $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.*

1. *(Inversion formula). For any $a < b$ we have*

$$\lim_{T\to\infty} \frac{1}{2\pi} \int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it} \phi_X(t)dt = \mathbb{P}^X((a,b)) + \frac{1}{2}\mathbb{P}^X(\{a\}) + \frac{1}{2}\mathbb{P}^X(\{b\}).$$

2. *(Uniqueness theorem). Assume that Y is another random variable such that $\phi_X(t) = \phi_Y(t)$. Then, X, Y have the same distribution $\mathbb{P}^X = \mathbb{P}^Y$.*

Example 1
Let $X \sim N(\mu_1, \sigma_1^1)$ and $Y \sim N(\mu_2, \sigma_2^2)$. We assume that X, Y are independent. Find the characteristic function of $Z = X + Y$ and the distribution of Z.

We have

$$\phi_Z(t) = \phi_{X+Y}(t) = \phi_X(t)\phi_Y(t) = e^{-t\mu_1 - t^2\frac{\sigma_1^2}{2}} e^{-t\mu_2 - t^2\frac{\sigma_2^2}{2}} = e^{-t(\mu_1+\mu_2) - t^2\frac{\sigma_1^2+\sigma_2^2}{2}}.$$

This is the characteristic function of a random variable with $N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$, so $Z \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$.

A corollary of the above theorem is the following.

Lemma 19.7. *Assume that X is random variable such that $\int_{\mathbb{R}} |\phi_X(t)|dt < \infty$. Then, the distribution of X has density*

$$f_X(x) := \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \phi_X(t)dt, \quad \text{for any } x \in \mathbb{R},$$

which is continuous and bounded.

19.2.2 Characteristic functions vs Moment generating functions

At a formal level, we see that if we set $t \mapsto it$ in the formula for a moment generating function $M_X(t) = \mathbb{E}[e^{tX}]$, then we get the characteristic function $\mathbb{E}[e^{itX}] = \phi_X(t)$. The formal equality

$$M_X(it) = \phi_X(t) \tag{19.1}$$

is true as long as M_X satisfies certain conditions so that complex inputs are allowed.

Theorem 19.8. *Let X be a random variable such that there exists $\varepsilon > 0$ such that $M_X(t) < \infty$ for every $t \in (-\varepsilon, \varepsilon)$. We consider the set*

$$A_\varepsilon = \{z \in \mathbb{C} : \ |Re(z)| < \varepsilon\}.$$

Then, M_X can be extended to a unique analytic function $h\colon A_\varepsilon \to \mathbb{C}$ such that

- *$h(t) = M_X(t)$, for any $t \in (-\varepsilon, \varepsilon)$.*
- *$h(it) = \phi_X(t)$.*

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Remark 19.9. The proof of the above theorem requires arguments from complex analysis. We may define $h\colon A_\varepsilon \to \mathbb{C}$ as $h(z) = \mathbb{E}[e^{zX}]$.

Using this theorem, we can now prove Theorem 13.4.

Proof of Theorem 13.4. Let h_X, h_Y be the functions provided by Theorem 19.8 associated with the random variables X, Y. Since $M_X = M_Y$ in $(-\varepsilon, \varepsilon)$, we have $h_X = h_Y$ in A_ε, therefore $\phi_X = \phi_Y$ in $\mathbb{R}$. The result follows from Theorem 19.6. $\square$

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